<table>
<tr><td></td><td></td><td>**[1], ans**</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>**8(a)(i)**</td><td>1. Any gas is made up of very **large number** of molecules.
2. The gas molecules are in **continuous random motion.**
3. The **volume** of the **gas molecules** is **negligible** compared to the volume of the container.
4. There are **no intermolecular forces** between molecules except during collisions between molecules.
5. The collision between gas molecules and between gas molecules and the wall of the container is **elastic.**
6. The **duration of collisions** is **negligible** compared with the time interval between collisions.</td><td>**[1]**
[1]
Any 2</td></tr>
<tr><td>**(ii)1.**</td><td>$E = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(290)$
$= 6.00 \times 10^{-21}$ J</td><td>[1] sub
[1] ans</td></tr>
<tr><td>**2.**</td><td>**It is possible** because **their molecular masses are different**</td><td>**[1]**</td></tr>
<tr><td>**(b)(i)**</td><td>**Internal energy** of a system is the **sum** of a **random distribution of kinetic and potential energies** associated with the **molecules** of the system.</td><td></td></tr>
<tr><td>**(ii)**</td><td><table>
<tr><td>Water freezing at constant temperature</td><td></td></tr>
<tr><td>A stone falling under gravity in a vacuum</td><td></td></tr>
<tr><td>Water evaporating at constant temperature</td><td>√</td></tr>
<tr><td>Stretching a wire at constant temperature.</td><td>√</td></tr>
</table></td><td>**[-1] for each error**</td></tr>
<tr><td>**(iii)**</td><td>The **random kinetic energy of the gas molecules increases** as the temperature increases.
The potential energy of the molecules remains unchanged since the volume is constant. → **Internal energy increases**</td><td>**[1]**
[1]</td></tr>
<tr><td>**(iv)1.**</td><td>W = $1.03 \times 10^{5}(1.87 \times 10^{-5} - 2.96 \times 10^{-2})$
= – 3050 J</td><td>**[1] sub**
[1] ans</td></tr>
<tr><td>**2.**</td><td>$U = Q + W$
$= 4.05 \times 10^{4} - 3050$
$= 3.75 \times 10^{4}$ J</td><td>[1] sub
[1] ans</td></tr>
</table>

JURONG JUNIOR COLLEGE
PHYSICS DEPARTMENT
2018 JC2 Preliminary Examinations
9749 H2 Physics Paper 3
Suggested Solutions with Markers' Comments

Qn	Suggested solution	Remarks
(c)(i)	• Two bodies of different masses at same temperature → same material would have different amount of internal energy • When a substance undergoes a change of state → the temperature remains unchanged although thermal energy is supplied. Temperature shows the direction of heat transfer → always from high to low temperature	**[1]** **[1]** **Any 2**
(ii)1.	To make allowance for heat gain from the atmosphere	**[1]**
2.	**Constant mass of water collected per minute in the beaker**	**[1]**
3.	mass melted by heater in 5 minutes = 64.7 – 8.3 = 56.4 g 56.4×10^{-3} (L) = 18 L = **319 kJ kg $^{-1}$**	 [1] sub [1] ans
MC	Use of m = 64.7, giving L = 278 kJ kg^{-1} no marks; Use of m = 48.1, giving L = 374 kJ kg^{-1} 1 mark	
9(a)	**Magnetic flux density** is defined as the **magnetic force per unit length per**	**[2]**